

RAHMAN HUSSAIN

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EDUCATION

Master of Science in Computer Science and Engineering
University at Buffalo, SUNY | GPA: 3.67

Jan 2025 – May 2026

WORK EXPERIENCE

Research Aide | SUNY Research Foundation
Apr 2026 – May 2026

- Built a two-tier log anomaly detection pipeline processing 2M+ security events. Implemented LoRA fine-tuned SLMs for high-speed Tier-1 triage and built a PyTorch Focal Loss MLP using Llama-70B embeddings for Tier-2 root-cause diagnosis.
- Engineered an automated multi-agent evaluation harness using a Commander-Coder loop, a graph-based RAG system (Neo4j), and a Qwen-32B ReAct judge to benchmark pipeline throughput and drive architectural optimizations.

Software Engineering Intern | Youro (Stealth Startup)

Aug 2025 – Dec 2025

- Architected Spring Boot APIs and a TypeScript HTTP client featuring in-flight request deduplication and TTL-based caching, eliminating mobile data over-fetching and cutting redundant backend calls by **30%**.
- Implemented comprehensive unit and integration testing suites using JUnit and Jest, establishing strict coverage baselines to prevent regression defects and stabilize CI/CD pipeline deployments.
- Resolved JVM OutOfMemory crashes during large patient data exports by replacing in-memory aggregations with chunked database streaming, processing records in batches of 1,000 and stabilizing memory consumption during peak utilization.

Software Developer | Goodz Pvt. Ltd

Jul 2023 – Dec 2024

- Migrated a legacy PHP/MySQL platform to an Angular SPA with reusable component architecture, hosting static assets on AWS S3 with CloudFront CDN and dropping page load times from **8 seconds to under 2 seconds**.
- Built a high-performance vendor dashboard using Angular CDK Virtual Scroll to render order records with row pinning and multi-column filtering, eliminating browser crashes and cutting DOM render time by **90%**.
- Architected reusable Angular service layers using dependency injection to share state and eliminate code duplication across the order management module.

PROJECTS

Image Indexing App | React, TypeScript, Python, FastAPI, Docker

2026

- Containerized a Python/FastAPI ML backend to serve a React frontend via Starlette, packaging the stack into a unified local asset; Authored Linux shell scripts to automate build caching and code-signing for the Tauri desktop installer.
- Optimized server-side logic by caching a single InsightFace ONNX model instance, eliminating C++ runtime memory crashes during parallel RESTful API requests.
- Engineered a React.js frontend to interface with local REST APIs, implementing efficient state management to render 10,000+ image metadata nodes without blocking the main UI thread.

Venyx Multi-Agent System | Python, FastAPI, Docker, API Development

2026

- Designed a hierarchical Commander-Worker agentic system using Claude and Gemini 2.5 Flash via MCP, coordinating 4+ specialized sub-agents across web search and Notion database tools, generating complete PRD documents in **under 15 seconds**.
- Architected a Python-based orchestration layer with structured prompt contracts and inter-agent communication protocols, enabling deterministic handoffs between 4+ specialized sub-agents without state bleed.

HealthAI Connect | Python | LangChain | Multi-modal AI | Computer Vision

2025

- Built and deployed a multi-modal ML pipeline using LangChain to chain retrieval and generation steps across vision and text models, producing structured clinical assessments. **Published in Taylor & Francis AI Frontiers**.
- Designed prompt workflows that anchored LLM outputs to structured medical context, cutting hallucination rate by grounding responses in patient-specific data across multi-modal inputs.

SKILLS

- **Languages:** Python, Java, TypeScript, JavaScript, SQL, C/C++
- **AI & Machine Learning:** PyTorch, LangChain, FAISS, LoRA, Matryoshka Representation Learning (MRL), LLM Integration
- **Frontend:** React, Angular, React Native, Tailwind CSS, HTML, CSS
- **Backend & APIs:** Spring Boot, Node.js, FastAPI, Flask, REST, WebSockets, GraphQL, SQLAlchemy
- **Infrastructure & Cloud:** Docker, AWS (S3, CloudFront), Kafka, CI/CD, GitHub Actions, Linux
- **Databases:** PostgreSQL, MySQL, Redis, Neo4j, Firebase
- **Testing & Tooling:** Jest, pytest, JUnit, ESLint, Husky, Maven, Vite, Git